

# Building a Retrieval-Augmented Generation (RAG) System for Course Material Question Answering

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## Problem & Motivation

Modern course materials, including lecture notes, slides, and assignments, are growing in volume and diversity. Traditional large language models (LLMs) may not be able to dynamically access this domain-specific content, resulting in incomplete or inaccurate answers. This project addresses the challenge of integrating external course materials into the response generation process via a Retrieval-Augmented Generation (RAG) framework. The approach is critical for enhancing answer accuracy in educational contexts, while also serving as a proof-of-concept for integrating dynamic, domain-specific knowledge into LLM outputs.

## Objectives

1. **Core Goal:** Develop a minimal viable prototype that preprocesses, indexes, and retrieves small-scale course materials, then integrates these with an LLM (GPT-4o) to generate informed answers.
2. **Evaluation:** Assess the retrieval effectiveness and answer relevance using basic quantitative metrics and manual review.
3. **Stretch Goal:** Conduct ablation studies to compare system performance using different LLMs and source materials

## Literature Review

- **Lewis et al. (2021)** *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks* [4]: This paper introduced the original RAG model, which combines a pre-trained generator (like BART) with a document retriever to answer questions using external information. They proposed two versions—one that uses the same retrieved documents for the whole answer and one that can use different documents for each word. RAG outperformed previous models on several open-domain QA benchmarks and showed that combining retrieval with generation leads to more accurate and specific responses
- **Cohere Documentation**[1]: Provides an overview of RAG architecture and its advantages in enhancing LLM performance. Offers detailed insights into implementing RAG, including methodologies and best practices.

- **Levonian et al. (2023)** *Retrieval-augmented Generation to Improve Math Question-Answering: Trade-offs Between Groundedness and Human Preference* [3]: This paper explores the challenges of domain- and context-specific question answering of LLMs. In this project, Levonian et al. focused on middle-school math student question answering that is appropriate to specific educational sources. They utilized RAG to incorporate external knowledge sources and found that students preferred the responses generated using RAG. This study shows the potential of RAG in the context of course material question answering.
- **Dong et al. (2023)** *How to Build an Adaptive AI Tutor for Any Course Using Knowledge Graph-Enhanced Retrieval-Augmented Generation (KG-RAG)* [2]: This paper proposes KG-RAG, which aims to solve the problems of LLM hallucinations and conceptual incoherence present in current RAG methods. By integrating a structured knowledge graph, Dong et al. improve the question-answering abilities of LLMs. The results show significant learning outcome improvements in both quantitative and qualitative assessments. The paper demonstrates the effectiveness of RAG approaches as well as potential for further improvements.

## Methodology

### Data

- **Source:** A limited set of course materials (e.g., the lecture notes and video captions, and potentially some textbooks).
- **Preprocessing:** Convert documents to plain text, clean the data, and implement a conservative chunking strategy to divide texts into semantically coherent segments.

### Approach

- **Embedding:** Use a pre-trained embedding model (e.g., Sentence Transformers) to generate vector representations for each text chunk.
- **Indexing:** Build a FAISS index for efficient similarity search.
- **Retrieval:** For a given user query, compute its embedding and retrieve the top- $k$  relevant chunks using a basic similarity search.
- **Prompt Construction:** Combine the query with retrieved chunks to build a context-rich prompt for the LLM.
- **Answer Generation:** Use an API call to get the LLM’s response.

### Evaluation

- **Sample Query Creation:** We will manually create ten to twenty questions for the model evaluation.
- **Baseline Comparison:** Compare the prototype’s performance against a system without retrieval augmentation.
- **LLM Ablation:** Evaluate performance with different LLMs (e.g., GPT-3 vs. GPT-3.5 and GPT-4o) to examine whether smaller models benefit more from RAG due to limited general knowledge.

- **Material Ablation:** Compare system performance using only lecture notes and captions vs. including textbook content to assess the marginal value of textbook information.

## Feasibility

Given the project timeline, the project scope is limited to a small-scale prototype:

- **Data:** Use a manageable dataset consisting of a few sample documents.
- **Compute:** Leverage pre-trained models and open-source libraries (e.g., FAISS, Sentence Transformers) to minimize computational overhead.
- **Time:** Prioritize core functionalities first (data processing, indexing, retrieval, and LLM integration) while reserving time for stretch goals if feasible.

## Expected Outcomes

- **Core Outcome:** A functional prototype demonstrating improved answer generation through retrieval-augmented prompts.
- **Stretch Outcomes:** Insights into the value of textbook content and LLM model size when using RAG.
- **Validation:** Results will be validated manually and presented in a table. If time permits, comparison of system variants and basic ablation studies (e.g., with vs. without the textbooks, and different LLMs) will be conducted to assess the impact of individual components.

## References

- [1] Cohere Documentation. Retrieval augmented generation (rag), 2025. <https://docs.cohere.com/v2/docs/retrieval-augmented-generation-rag>.
- [2] Chenxi Dong, Yimin Yuan, Kan Chen, Shupeí Cheng, and Chujie Wen. How to build an adaptive ai tutor for any course using knowledge graph-enhanced retrieval-augmented generation (kg-rag), 2025.
- [3] Zachary Levonian, Chenglu Li, Wangda Zhu, Anoushka Gade, Owen Henkel, Millie-Ellen Postle, and Wanli Xing. Retrieval-augmented generation to improve math question-answering: Trade-offs between groundedness and human preference, 2023.
- [4] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021. <https://arxiv.org/abs/2005.11401>.